HNS Projekt 2

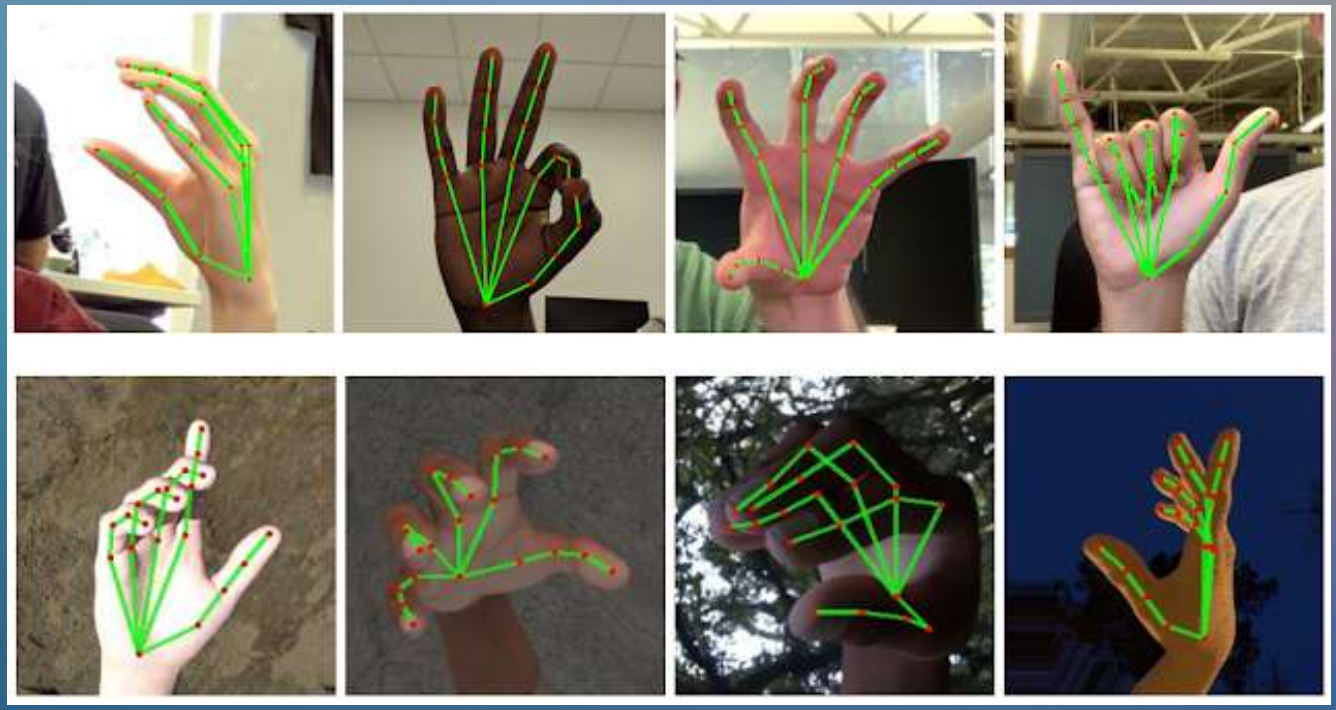
Michal Čopjan
Klára Murínová

Rozpoznávanie gest amerického posunkového jazyka na kamere

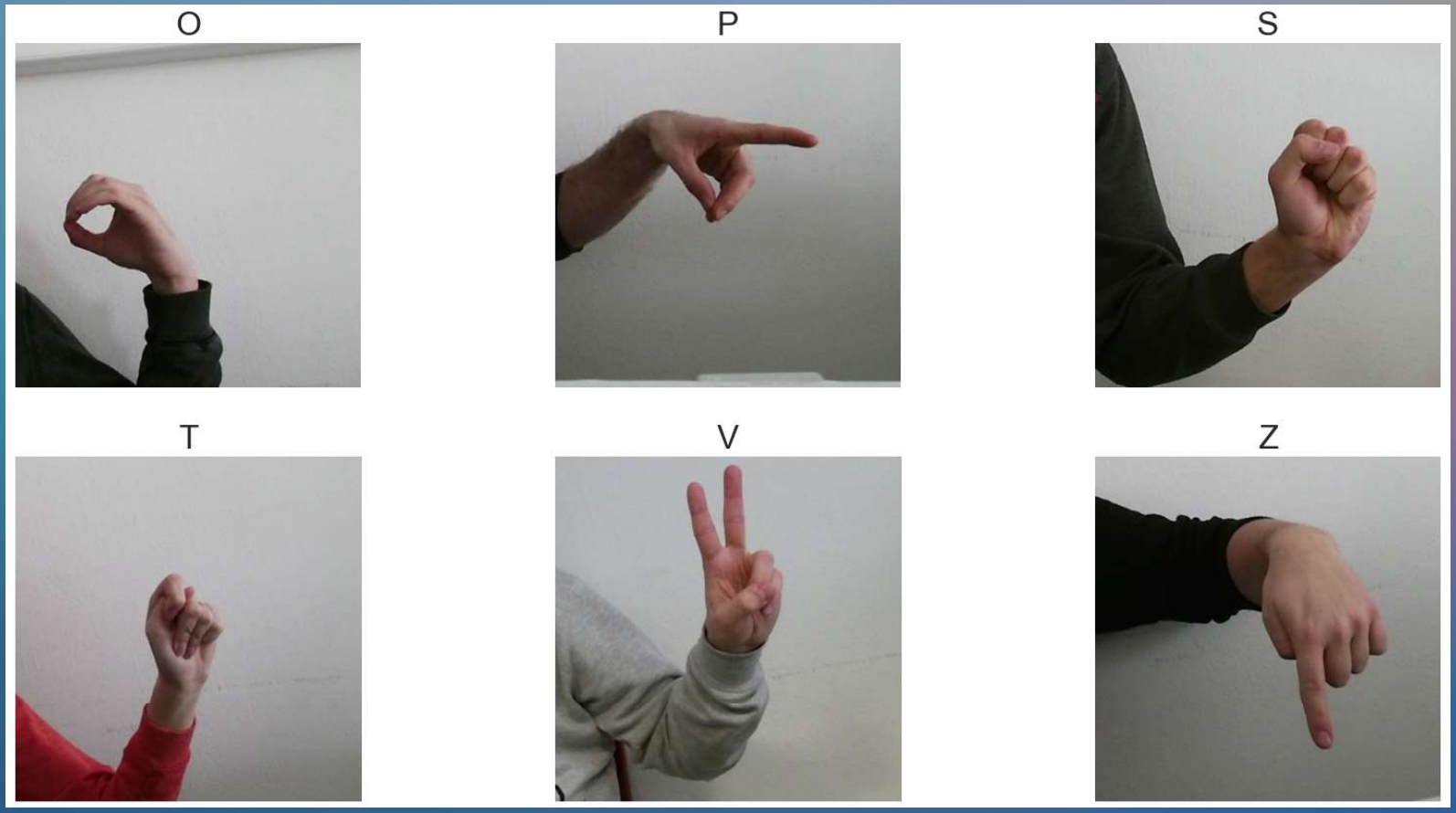
Flow


Zachytenie snímky z kamery
Detekcia ruky
Klasifikovanie gesta
V
Nakreslenie bboxu a triedy na snímku
Zobrazenie snímky
V

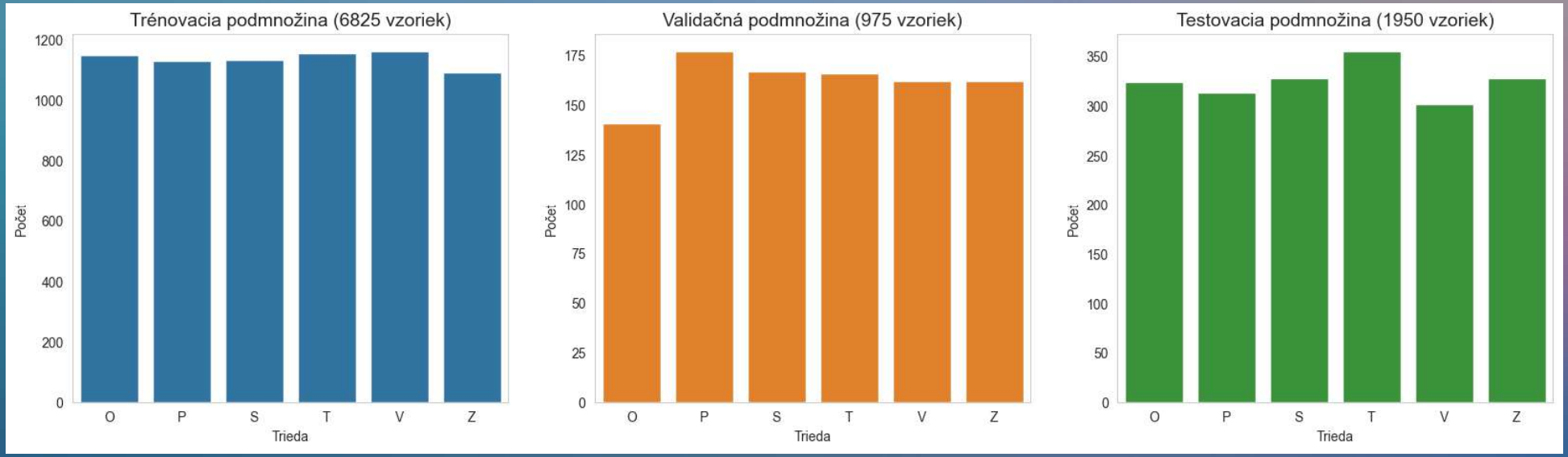
Google MediaPipe detektor rúk



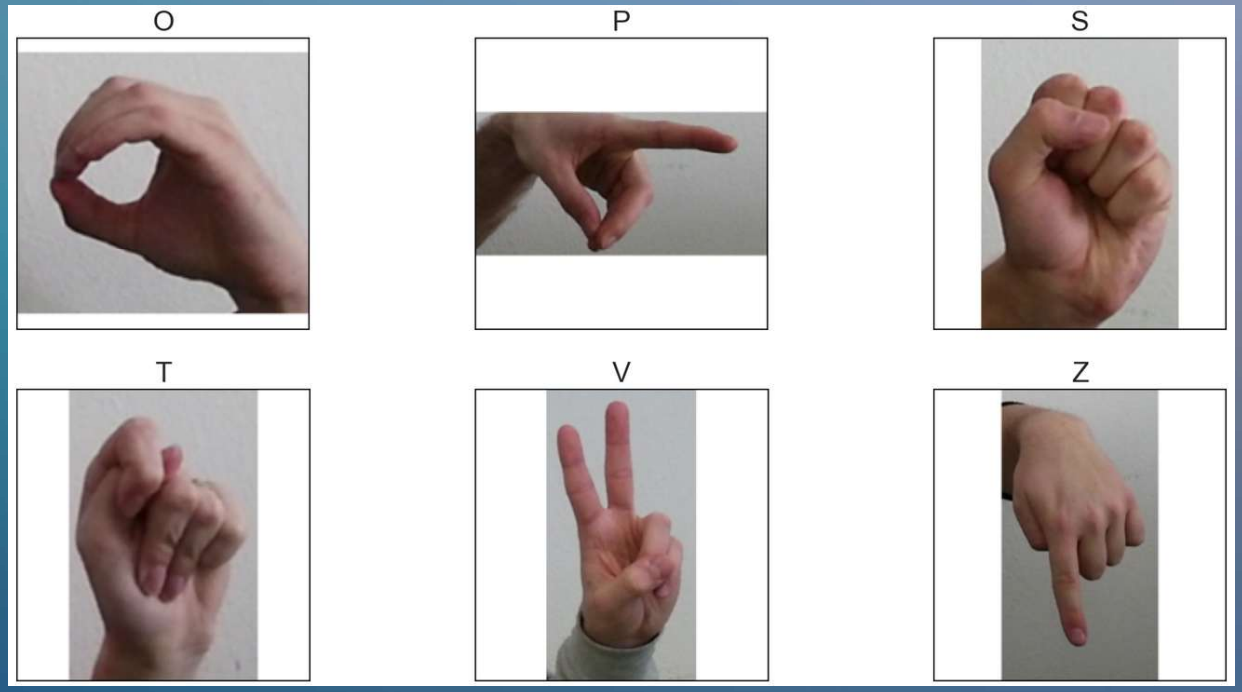
Dataset



Dataset


Trénovacia podmnožina (6825 vzoriek)
Počet
1200
1000
800
600
400
200
0
O
P
S
T
V
Z
Trieda
Validačná podmnožina (975 vzoriek)
Počet
175
150
125
100
75
50
25
0
O
P
S
T
V
Z
Trieda
Testovacia podmnožina (1950 vzoriek)
Počet
350
300
250
200
150
100
50
0
O
P
S
T
V
Z
Trieda

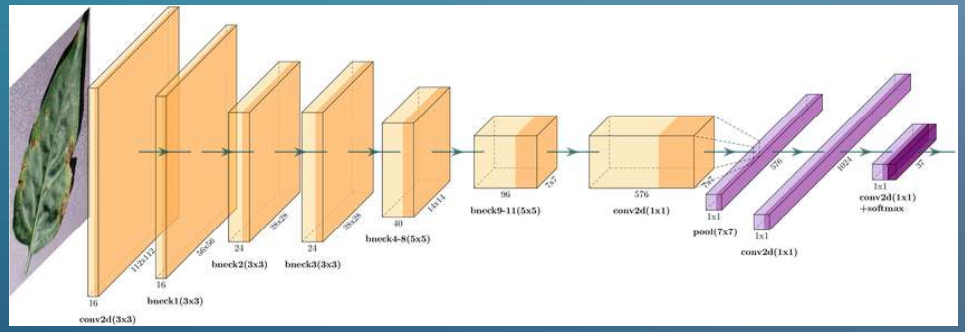
Dataset




```
transforms.RandomAffine(
    degrees=8,
    translate=(0.05, 0.08),
    shear=2
),
transforms.ColorJitter(
    brightness=0.15,
    contrast=0.15,
    saturation=0.10,
    hue=0.02
),
```

Klasifikátory

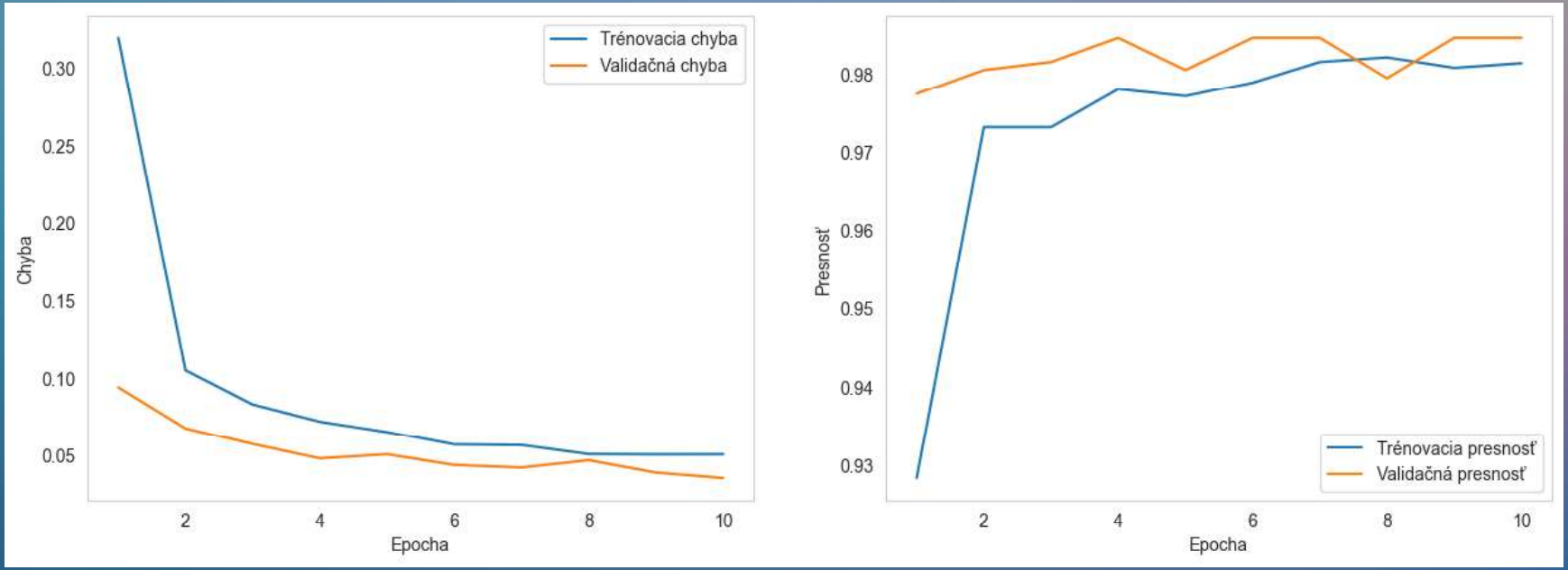
MobileNetV3-Small



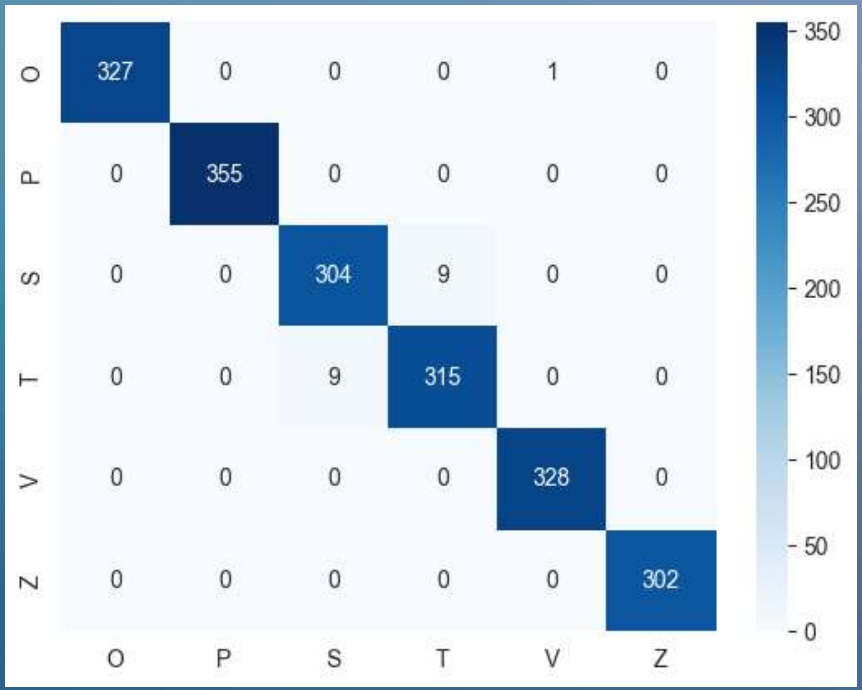
Vlastná CNN

```
self.conv1 = nn.Conv2d(in_channels=3, out_channels=16, kernel_size=3, padding=1)
self.bn1 = nn.BatchNorm2d(16)
self.conv2 = nn.Conv2d(in_channels=16, out_channels=32, kernel_size=3, padding=1)
self.bn2 = nn.BatchNorm2d(32)
self.conv3 = nn.Conv2d(in_channels=32, out_channels=64, kernel_size=3, padding=1)
self.bn3 = nn.BatchNorm2d(64)
self.pool = nn.MaxPool2d(2, 2)
self.adaptive_pool = nn.AdaptiveAvgPool2d((1, 1))
self.dropout = nn.Dropout(0.5)
self.fc1 = nn.Linear(64, 256)
self.fc2 = nn.Linear(256, NUM_OF_CLASSES)
```

MobileNetV3-Small



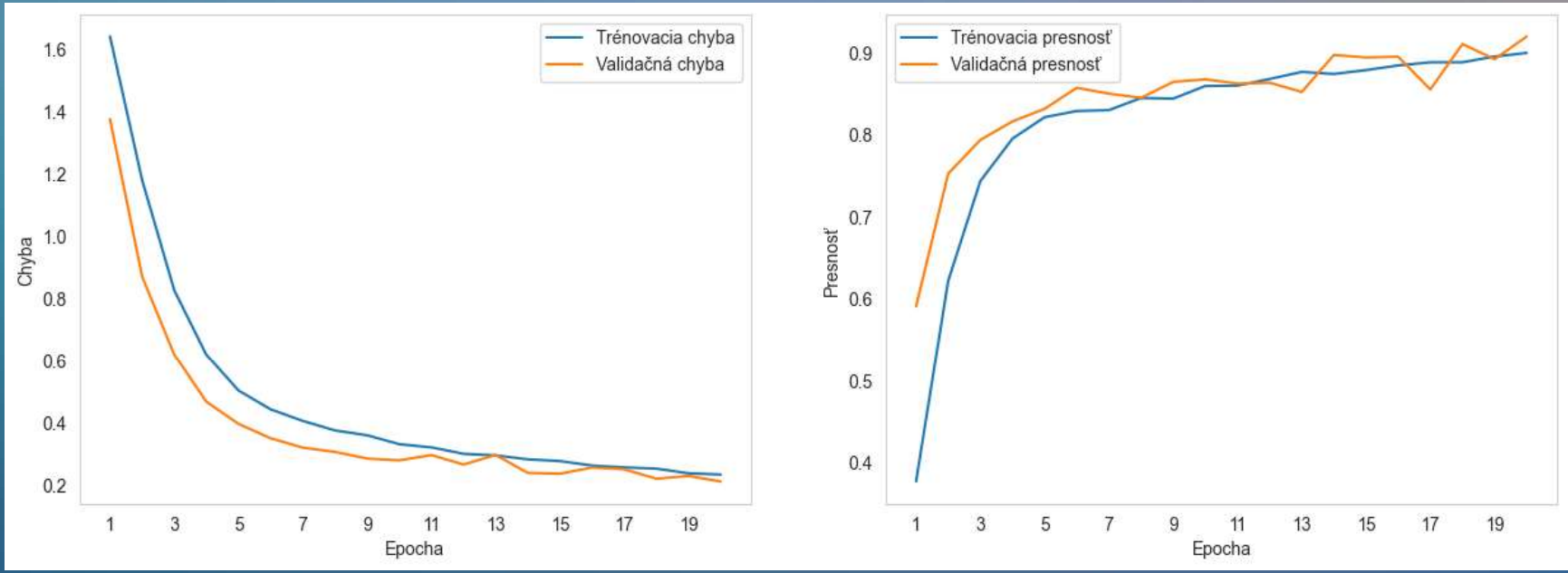
MobileNetV3-Small



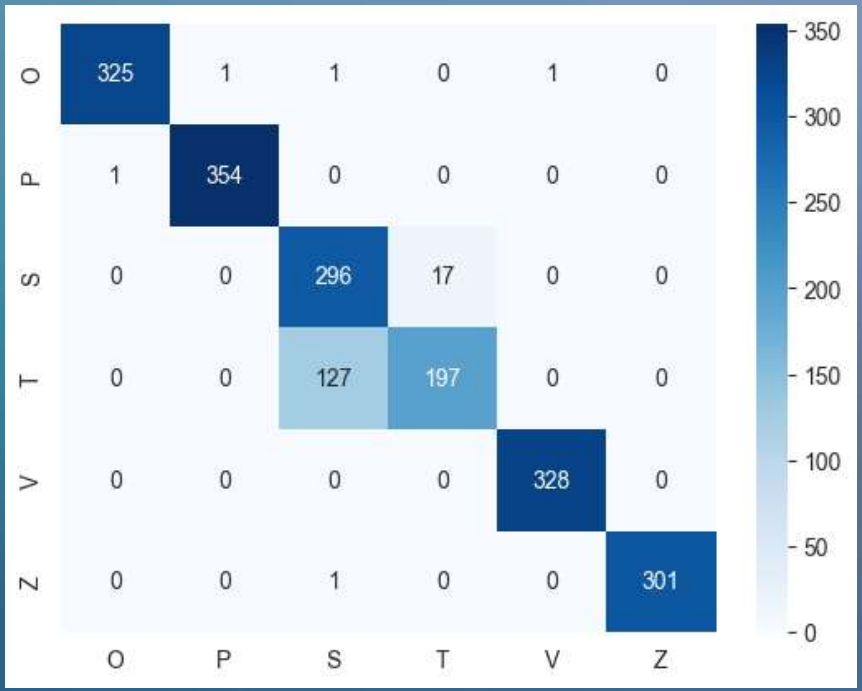
99.03 %

Presnosť

Vlastná CNN


Trénovacia chyba
Validačná chyba
Chyba
Epocha
Trénovacia presnosť
Validačná presnosť
Presnosť
Epocha

Vlastná CNN



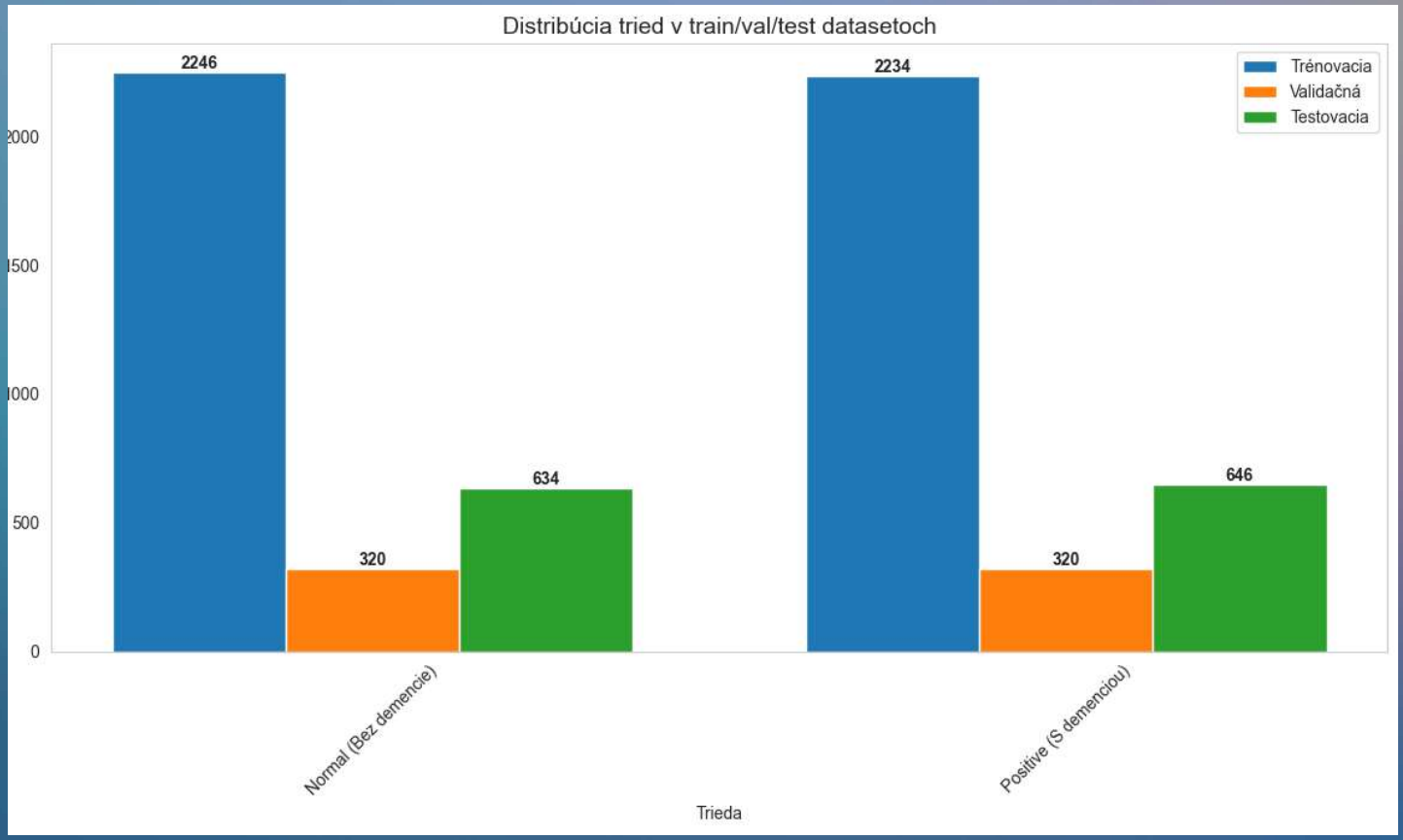
92.36 %

Presnosť

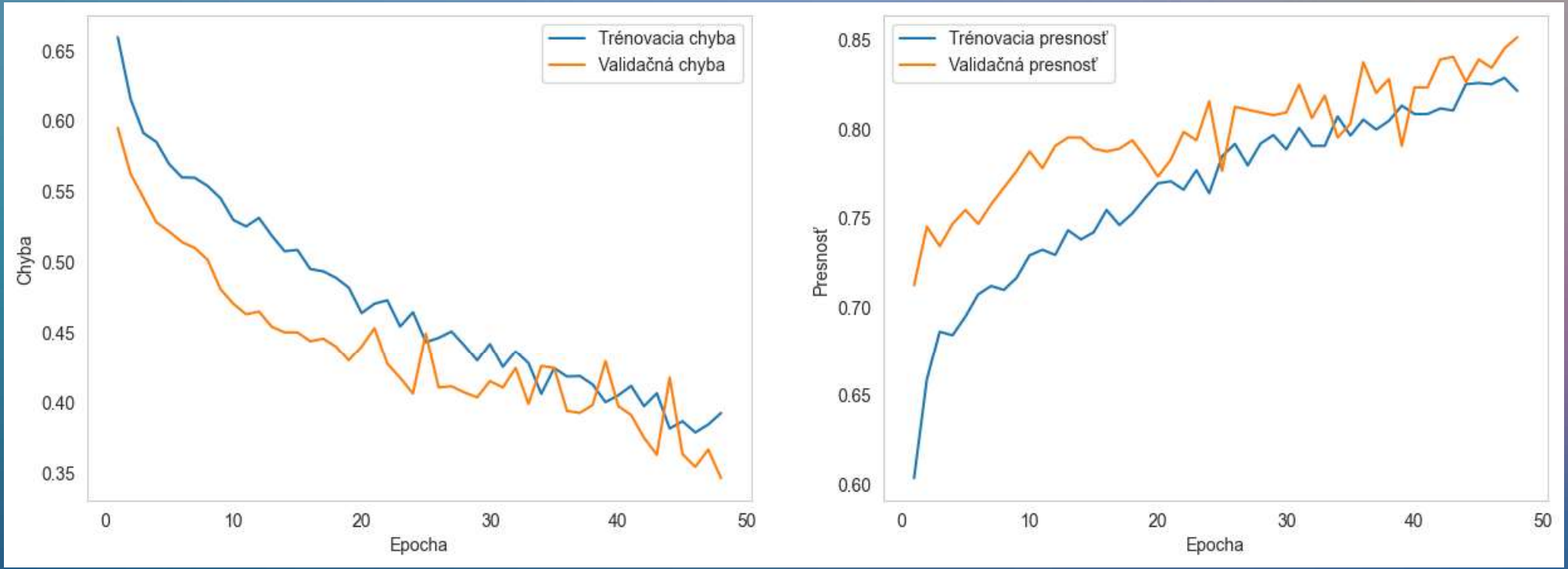
Ukážka

Rozpoznávanie Alzheimerovej choroby z MRI snímkov

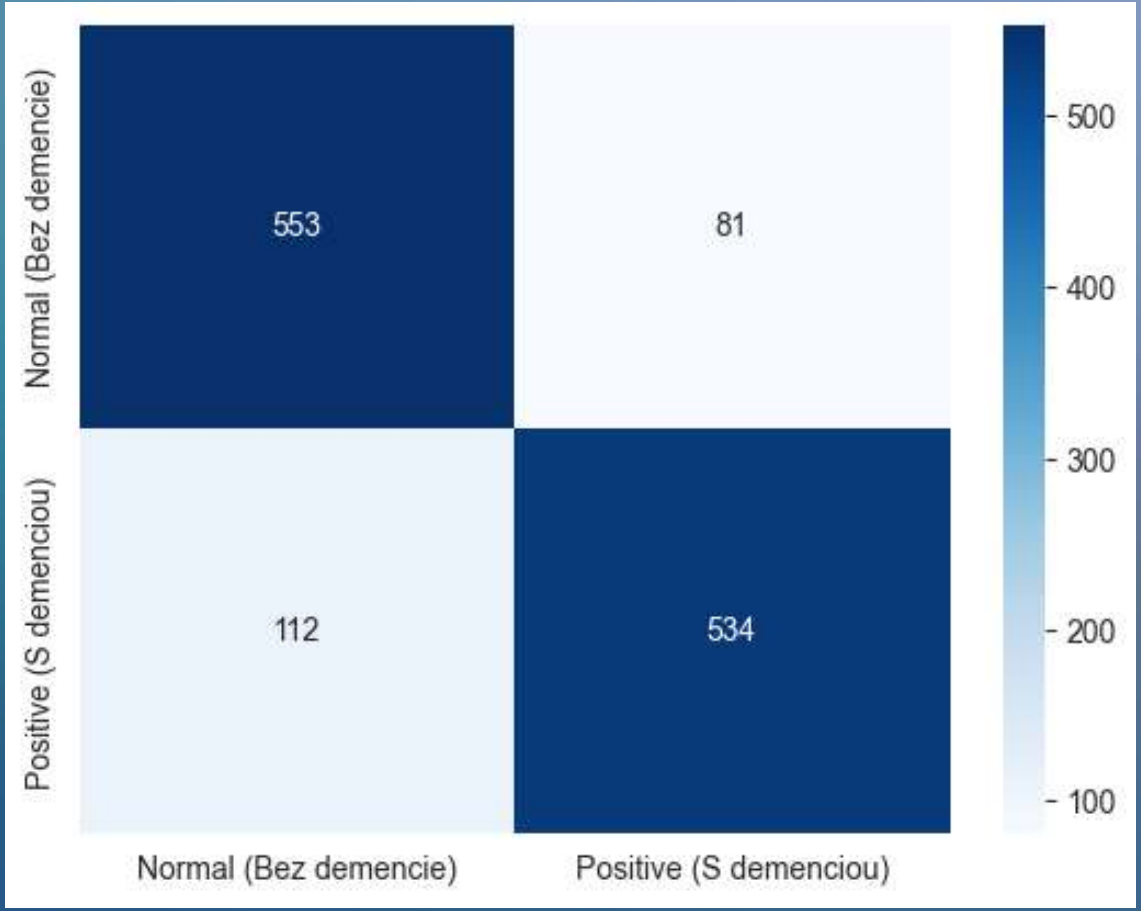
Dataset


Distribúcia tried v train/val/test datasetoch
Trénovacia
Validačná
Testovacia
2246
320
634
2234
320
646
2000
1500
1000
500
0
Normal (Bez demencie)
Positive (S demenciou)
Trieda

EfficienteNet B3



EfficienteNet B3



84.92%
Presnosť

82.66%
Senzitivita

87.22%
Špecificita